Corporate Finance and Financial Modelling: Term Project

Predicting Financial Distress Using Altman's Z-Score Model: A Matched-Pair Study of Distressed and Healthy Indian Listed Non-Financial Companies

Area: Investment (financial-health and default-risk assessment before investment decisions)

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Base paper

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Abstract

This study tests whether Altman's (1968) Z-score would have warned investors about Indian listed companies that were later admitted to insolvency under the Insolvency and Bankruptcy Code, 2016. Fifteen non-financial companies admitted to the Corporate Insolvency Resolution Process between July 2017 and June 2024 are matched with fifteen healthy listed peers from the same industries, and Altman's five ratios are computed for the three fiscal years before each admission (85 firm-years). Standalone financial data from two independent compilations agree to within 0.01% at the median, and current assets and liabilities are taken from reported statements. Paired t-tests and non-parametric tests show that all five ratios are significantly lower for the distressed firms one year before the event, with the largest effects for profitability (X3), retained earnings (X2) and working capital (X1). Every distressed company was already in Altman's distress zone ($Z < 1.81$) one year before admission, so the Type I error was zero, but 4 of 15 healthy peers were also below 1.81, giving overall accuracy of 87% (80% with the 2.675 cut-off). The emerging-market Z''-score classified 97% correctly. Discriminating power stays high over time (AUC 1.00, 0.98 and 0.91 at $t-1$, $t-2$ and $t-3$), and the score flagged 11 of 13 distressed firms three years ahead. A discriminant function re-estimated on Indian data ranks retained earnings and profitability as the most important ratios, and a cut-off of about 1.17 fits Indian firms better than 1.81. Between the end of year $t-1$ and the admission date, distressed shares lost a median 25% while their peers gained 10%. The Z-score is therefore a reliable, low-cost first screen for Indian equity and credit investors, provided it is applied with industry-aware cut-offs.

Keywords: Altman Z-score; financial distress; Insolvency and Bankruptcy Code; discriminant analysis; logistic regression; India; investment screening.

1. Introduction

An equity or debt investor who commits capital to a company is exposed to the possibility that the company will be unable to meet its obligations. When that happens, shareholders are usually wiped out and creditors recover only part of their claims, so an early and inexpensive signal of deteriorating financial health is valuable at the screening stage of any investment decision. Accounting-based distress models turn the information in published financial statements into such a signal. The best known of these is the Z-score of Altman (1968), which combines five financial ratios through multiple discriminant analysis (MDA) into a single score that separates firms likely to fail from firms likely to survive.

India offers a natural test bed for the model. Before 2016 a creditor of a defaulting Indian company had to work through a fragmented set of laws (the Sick Industrial Companies Act, debt recovery tribunals, the SARFAESI Act and winding-up petitions in High Courts), and large defaults were often not resolved for years. The Insolvency and Bankruptcy Code, 2016 (IBC) replaced this with a single time-bound process: once the National Company Law Tribunal (NCLT) admits an application, a Corporate Insolvency Resolution Process (CIRP) starts, the board is suspended and an insolvency professional takes control. On 13 June 2017 the Reserve Bank of India (RBI) directed banks to refer twelve large accounts, which together made up about a quarter of banking-system non-performing assets, for insolvency under the Code (Reserve Bank of India, 2017). Many of them were listed companies whose financial statements had been public for years. The admission of these and later cases gives a clearly dated, legally defined distress event for a group of listed Indian firms, which is close in spirit to the bankruptcy-petition event that Altman used for US manufacturers.

This project applies Altman's original Z-score and its emerging-market variant (the Z"-score) to fifteen Indian listed non-financial companies admitted to CIRP between 2017 and 2024, each matched with a healthy listed company from the same industry, and measures how well the scores would have warned an investor one, two and three years before the event. It also re-estimates the discriminant function and a logistic model on the Indian data, to show which ratios matter most for Indian firms and whether Altman's 1968 weights and cut-offs still hold.

2. Literature Review

Univariate and multivariate beginnings. Beaver (1966) showed that individual ratios, especially cash flow to total debt, differ between failed and non-failed US firms up to five years before failure. Altman (1968) argued that single ratios can give conflicting signals and used MDA on 33 bankrupt and 33 matched non-bankrupt manufacturers (1946–1965) to combine working capital/total assets, retained earnings/total assets, EBIT/total assets, market value of equity/book value of total liabilities and sales/total assets. The resulting function classified 95% of the original sample correctly one year before bankruptcy (Type I error 6%, Type II error 3%), but accuracy for bankrupt firms fell to 72% two years before and to about half three years before. Altman reported a cut-off of 2.675 that minimised misclassifications and a 'zone of ignorance' between 1.81 and 2.99. Deakin (1972) applied the fourteen ratios that Beaver had examined one at a time within a series of multivariate discriminant models.

Refinements and alternative estimators. Altman, Haldeman and Narayanan (1977) built the seven-variable ZETA model on a larger and more recent sample and improved accuracy at

longer horizons. Taffler (1983) developed a UK Z-score for listed manufacturers. Ohlson (1980) replaced MDA with logistic regression, which does not need multivariate normality or equal covariance matrices and gives a probability of failure directly. Zmijewski (1984) showed that choice-based (matched) samples and the exclusion of firms with incomplete data bias estimated coefficients and overstate accuracy, a caution that applies to every matched-pair design, including this one. Shumway (2001) argued that single-period models ignore how risk changes over time and proposed a hazard model that also adds market variables.

Out-of-sample stability. Begley, Ming and Watts (1996) found that the original Altman and Ohlson models perform worse on 1980s data than in the periods in which they were estimated, even when their coefficients are re-estimated. Grice and Ingram (2001) found that the original Z-score predicts bankruptcy less accurately in recent samples than in Altman's original periods and is less accurate for non-manufacturing firms, although it remains useful for identifying financial stress short of bankruptcy. Altman, Iwanicz-Drozdowska, Laitinen and Suvas (2017) tested the Z"-score on firms from 34 countries and concluded that the model, even with its original coefficients, keeps useful classification power internationally, and that re-estimating it on local data and adding country- or industry-specific information improves accuracy. Altman (2018) reviews fifty years of applications of the Z-score family, including the Z" version for non-manufacturers and emerging markets, which omits the asset-turnover ratio (X5).

Accounting versus market information. Hillegeist, Keating, Cram and Lundstedt (2004) found that a Black-Scholes-Merton probability of bankruptcy carries more information than the Z-score or O-score, whereas Agarwal and Taffler (2008) found that for UK firms the accounting-based z-score performs as well as market-based models and gives larger economic benefits when used for credit pricing. Campbell, Hilscher and Szilagyi (2008) linked distress to equity returns, showing that stocks with high predicted failure risk earn anomalously low subsequent returns. That result gives the investor in our study a direct reason to screen for distress before buying. Wu, Gaunt and Gray (2010) compared five established models on up-to-date data and found that each contains unique information but that their performance varies over time. A combined model using key variables from all five, plus a diversification measure, outperformed them in-sample and out of sample.

Emerging-market and Indian evidence. Altman (2005) proposed the emerging-market (EM) score, the four-variable Z"-model with a constant of 3.25 added so that the score maps onto US bond-rating equivalents, as a starting point for assessing emerging-market corporate credit. In India, Bandyopadhyay (2006) estimated MDA and logit models for corporate-bond defaults and found that a re-estimated Z-score clearly beat the original Altman model in the Indian context.

Singh and Mishra (2016) re-estimated the Altman, Ohlson and Zmijewski models on 208 Indian manufacturing firms and showed that local re-estimation improves accuracy in both estimation and hold-out samples. Singh and Singla (2023) re-estimated the Z-score coefficients on recent Indian data and found, using ROC analysis on a test sample, that a newly built logistic model predicted distress more accurately. Most Indian studies define distress through rating downgrades, defaults on bonds or BIFR registration. Fewer use the legally defined IBC admission event, and fewer still examine the score's value at the investor's screening horizon of two to three years. This project addresses that gap.

Table 1. Summary of key studies reviewed

Study	Sample	Method	Main finding
Altman (1968)	US, 33+33 manufacturers	MDA, five ratios	95% accuracy at t−1; 72% of bankrupt firms at t−2
Ohlson (1980)	US industrial firms, 1970–76	Logit (O-score)	Probabilistic prediction; earlier studies may overstate accuracy
Zmijewski (1984)	US, NYSE/AMEX firms	Probit; bias analysis	Matched samples overstate accuracy
Grice & Ingram (2001)	US, recent samples	Original Z re-tested	Less accurate in recent data and outside manufacturing
Altman (2005)	Emerging-market bonds	EM / Z"-score	Four-ratio model without asset turnover
Bandyopadhyay (2006)	India, bond defaults	MDA and logit	Re-estimated Z beats original model
Singh & Mishra (2016)	India, 208 manufacturers	MDA, probit, logit	Re-estimation improves all three models
Altman et al. (2017)	34 countries	Z"-score, logit	Original model still useful; local re-estimation helps

3. Research Questions and Focus of the Work

The work focuses on one practical question: is Altman's Z-score, built in 1968 on US manufacturers, a reliable screening tool for an investor evaluating Indian listed companies today? This question is split into four research questions.

RQ1. Do the five Altman ratios (X1–X5), and the composite Z and Z" scores, differ significantly between Indian companies that were later admitted to insolvency and matched healthy companies one, two and three years before the event?

RQ2. How accurately do Altman's original Z-score (with the 1.81 / 2.675 / 2.99 cut-offs) and the emerging-market Z"-score (1.10 / 2.60) classify the two groups at each horizon, and how do these accuracies compare with Altman's 1968 benchmarks?

RQ3. When the discriminant function is re-estimated on Indian data, and when a logistic model is fitted, which ratios carry the most weight? Do the re-estimated models classify better than the original coefficients out of sample?

RQ4. What do the results imply for an investor who uses the Z-score as a screening tool, in terms of how early the warning comes, how wrong it can be, and what happens to the share price?

For each ratio and score, the null hypothesis is H0: μ(distressed) = μ(healthy); the alternative is H1: μ(distressed) ≠ μ(healthy). Based on Altman (1968), we expect the distressed group to have lower values of X1–X4 and of Z.

4. Dataset and Methodology

4.1 Sample design

Following Altman's paired-sample design, the study uses 15 distressed and 15 healthy companies listed on the NSE or BSE. A company is classed as distressed if the NCLT admitted it into CIRP under the IBC; the event date is the date of the admission order, confirmed from the company's stock-exchange filing, the IBBI order listing or contemporaneous press reports (Appendix). Banks, NBFCs and insurers are excluded because their balance sheets are not comparable with the Z-score ratios. Each distressed company is matched with a listed company from the same industry that never defaulted over the sample period, chosen as the closest in size among the industry's leading listed firms. Admissions span July 2017 to June 2024 and cover airlines, steel, textiles, real estate, power, auto components, EPC, telecom, edible oils, education, paper, retail and cement.

Timing is set so that the investor could actually have seen the statements. Year t−1 is the last fiscal year (ending 31 March) that closed at least 90 days before the admission order. SEBI's Listing Regulations require audited annual results within 60 days of the year end (Securities and Exchange Board of India, 2015, Reg. 33(3)(d)), and the extra 30 days allows for distressed firms that file late. Years t−2 and t−3 are the two preceding fiscal years. Each healthy peer is measured on exactly the same fiscal years as its distressed match, so macroeconomic conditions are held constant within each pair. One exception follows the same principle: Future Retail told the exchanges on 24 May 2022 that approval of its FY2022 results was delayed because its board and audit committee were not properly constituted, and no FY2022 statements appear in either data source, so its t−1 is FY2021, the latest year with audited statements available.

Several other well-known IBC cases were examined and excluded: Monnet Ispat & Energy Ltd (no traded share price for FY2015–FY2017 in the price history); Videocon Industries Ltd (irregular 18-, 12- and 15-month reporting periods before the event); Era Infra Engineering Ltd (no traded share price available for the pre-event years); ABG Shipyard Ltd (no listed shipbuilding peer with a share price for FY2015–FY2017); Reliance Naval and Engineering Ltd (historical statements not available in the data source); Electrosteel Steels Ltd; Uttam Galva Steels Ltd (historical statements not available in the data source after delisting); Essar Steel Ltd; Bhushan Power & Steel Ltd (not listed on an Indian exchange in the pre-event years).

4.2 Data sources and validation

All accounting figures are standalone, because the IBC process applies to the legal entity, and are in Rs crore. The income statement and balance sheet (revenue from operations, profit before tax, interest, reserves and surplus, borrowings, other liabilities and total assets) were taken from Screener.in, which compiles the statements that listed companies file with the exchanges. A second, independent compilation, LSEG (Refinitiv) standalone annual-report data published on Sharpely.in, was used to check them and to supply reported totals of current assets and current liabilities. Where LSEG data were missing, these totals were read directly from the standalone balance sheets in the company annual reports (Bhushan Steel, Bharat Forge, Sintex Industries and JK Paper) obtained from the BSE archive or the company's website.

The two compilations were reconciled before use. For the 67 firm-years covered by both, the median absolute difference is below 0.01% for total assets, revenue, profit before tax and reserves. 91% of total-asset figures, 96% of revenue figures and 97% of profit figures agree within 5%; the remaining gaps come from later restatements, such as the Ind AS transition. Current assets and liabilities come from LSEG for 67 of the 85 firm-years, from annual reports for 11, and from Sharpely's statement table for 7; the table is used only when its total assets agree with the standalone figure within 5%. X1 is computed with the total assets of the same source, which agree with Screener's at a median gap below 0.01% (91% within 5%). No working-capital figure is estimated. Retained earnings are measured by 'reserves and surplus' (other equity under Ind AS), and both compilations agree on this figure. It also includes securities premium, which is the usual limitation of Indian Z-score studies.

The market value of equity is the last NSE closing price on or before 31 March of each fiscal year multiplied by the number of shares. Because the price history is adjusted for later bonuses, splits and rights issues, the share count must be adjusted on the same basis. LSEG's basic weighted share count meets this requirement and is used for 62 firm-years; net profit divided by EPS is used for the other 23. As a check, Bharti Airtel's computed market value at 31 March 2017 (Rs 1,39,926 crore) matches the Rs 1,398.89 billion reported in its 2016-17 annual report to within 0.1%.

4.3 Variables

Table 2. Variable definitions

Code	Ratio	Measurement	Dimension
X1	Working capital / Total assets	(Current assets − Current liabilities) / Total assets	Liquidity
X2	Retained earnings / Total assets	Reserves and surplus / Total assets	Cumulative profitability, age
X3	EBIT / Total assets	(Profit before tax + Interest) / Total assets	Operating productivity

X4	MVE / Total liabilities	Year-end market capitalisation / (Borrowings + Other liabilities)	Solvency (market)
X4'	Book equity / Total liabilities	(Share capital + Reserves) / Total liabilities	Solvency (book), used in Z"
X5	Sales / Total assets	Revenue from operations / Total assets	Asset turnover

4.4 Models and statistical methods

(a) Original Z-score (Altman, 1968): $Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5$. Altman's zones are distress ($Z < 1.81$), grey ($1.81 \leq Z \leq 2.99$) and safe ($Z > 2.99$). A single cut-off of 2.675 minimised misclassification in his sample.

(b) Emerging-market Z"-score (Altman, 2005; Altman et al., 2019): $Z'' = 6.56X1 + 3.26X2 + 6.72X3 + 1.05X4'$, with book equity in X4' and no asset-turnover term. Its zones are distress ($Z'' < 1.10$), grey (1.10–2.60) and safe ($Z'' > 2.60$). Adding the constant 3.25 gives the EM score used for bond-rating equivalents. The constant does not change any classification.

(c) Ratio analysis and hypothesis tests. Each ratio is described by group and horizon. Because the design is matched, the main test of H0 is a paired t-test on the within-pair differences. A Welch two-sample t-test (unequal variances), the Wilcoxon signed-rank test and the Mann-Whitney U test are reported as robustness checks, since Shapiro-Wilk tests reject normality for several ratios. Effect sizes are Cohen's d. 95% confidence intervals for the mean within-pair difference are bootstrapped with 5,000 resamples.

(d) Re-estimated discriminant analysis. Altman's MDA is re-estimated on the Indian t−1 sample with the same five ratios. To stop a few extreme values from dominating the estimation, ratios are winsorised at the pooled 5th and 95th percentiles before estimation. The output reported is the unstandardised coefficients, Wilks' lambda with Bartlett's χ^2 test, the univariate F-ratio for each variable (as in Altman, 1968), and the scaled coefficients (coefficient × pooled standard deviation) used to rank each variable's contribution. Accuracy is measured in-sample, by leave-one-out cross-validation (Lachenbruch & Mickey, 1968), and on the t−2 and t−3 hold-out years.

(e) Logistic regression. Following Ohlson (1980), a logit of the distress dummy is estimated on Z alone (Model 1) and on X1–X5 (Model 2). When maximum likelihood fails because of complete separation, which is common when groups are well separated and samples are small, Firth's (1993) penalised likelihood is used. Coefficients, odds ratios, McFadden's pseudo-R^2, the likelihood-ratio test and AIC are reported.

(f) Model comparison. Discriminating power is summarised by the area under the ROC curve (AUC), with bootstrapped 95% confidence intervals. Differences between correlated AUCs are tested with the DeLong, DeLong and Clarke-Pearson (1988) test. For the re-estimated models, leave-one-out scores are used so that the comparison is out of sample. The Youden index

(sensitivity + specificity − 1) gives the cut-off that best fits the Indian sample. Sensitivity checks vary the X4 definition, the working-capital source and the industry composition.

(g) Investment view. For each distressed firm, the share-price change from the end of fiscal year t−1 to the last trading day before the admission order is compared with the change for its peer over the same window. This shows what a Z-score warning was worth to a shareholder.

All calculations were done in Python (pandas, SciPy, statsmodels) with a script that is submitted with the report. The Excel workbook contains the raw figures, every ratio and every test statistic, so each number in this report can be traced back to its source.

5. Results

5.1 Ratio analysis: descriptive statistics

Table 3. Mean (median) of the Altman ratios and scores by group and horizon

Variable	t−1 D	t−1 H	t−2 D	t−2 H	t−3 D	t−3 H
X1 – Working capital / Total assets	−0.32 (−0.32)	0.13 (0.16)	−0.09 (−0.06)	0.12 (0.14)	−0.01 (0.03)	0.10 (0.12)
X2 – Retained earnings / Total assets	0.03 (0.05)	0.45 (0.44)	0.12 (0.14)	0.46 (0.45)	0.20 (0.15)	0.44 (0.42)
X3 – EBIT / Total assets	−0.04 (−0.03)	0.10 (0.09)	−0.01 (0.01)	0.11 (0.11)	0.02 (0.03)	0.13 (0.11)
X4 – Market value of equity / Total liabilities	0.09 (0.05)	11.72 (1.65)	0.12 (0.07)	12.14 (1.69)	0.40 (0.13)	7.61 (1.09)
X5 – Sales / Total assets	0.36 (0.15)	0.93 (0.77)	0.44 (0.20)	0.79 (0.70)	0.53 (0.20)	0.94 (0.73)
Altman Z-score	−0.06 (−0.05)	9.09 (2.83)	0.55 (0.31)	9.20 (2.77)	1.09 (0.99)	6.68 (2.78)
Z''-score (emerging market)	−2.15 (−2.14)	5.12 (3.68)	−0.01 (0.19)	5.16 (3.56)	1.06 (1.13)	4.19 (3.71)

D = distressed (IBC) group; H = healthy peer group. n = 15 pairs at t−1, 15 distressed / 14 healthy at t−2, 13 / 13 at t−3. Raw (unwinsorised) values.

Table 3 shows a clear pattern on every dimension of financial health. One year before admission, the median distressed firm had negative working capital (X1 median −0.32), so its current liabilities exceeded current assets by about a third of total assets. This is partly because defaulted long-term loans had been reclassified as current once lenders recalled them. Retained earnings were close to zero (X2 median 0.05 against 0.44 for peers) after several years of losses, and operating returns were negative (X3 median −0.03 against 0.09). The market had already priced in the distress: the median market value of equity covered only 5% of book liabilities, against 1.6 times for the peers. Asset turnover (X5) was also lower, because distressed firms carried large unproductive assets such as stalled projects, idle plants and grounded aircraft. Healthy-group means for X4 and Z are pulled up by two very lightly leveraged peers (Avenue Supermarts and Aptech), so the medians are the better summary. The trend across the three years is also informative. For the distressed firms, the median Z falls from 0.99 at t−3 to 0.31 at t−2 and −0.05 at t−1, while the healthy median stays close to 2.8 (Figure 1). Among the

distressed firms, the median X1 falls from 0.03 to −0.32 and the median X3 from 0.03 to −0.03. Liquidity and operating profitability erode first, and accumulated losses then eat into retained earnings.

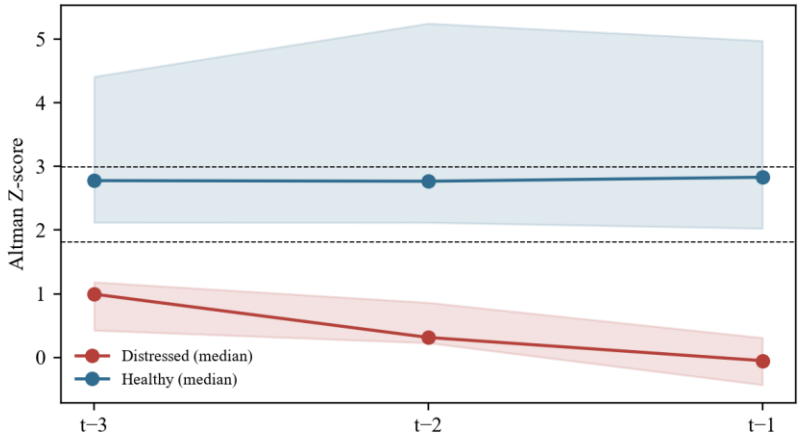


Figure 1. Median Z-score (shaded: inter-quartile range) from t−3 to t−1; dashed lines mark 1.81 and 2.99

Correlations among the winsorised t−1 ratios are positive and moderate to high (for example r(X2, X3) = 0.68 and r(X1, X3) = 0.75), as expected because all five ratios reflect the same underlying condition. Variance inflation factors range from 1.5 to 4.0, below the usual threshold of 5 to 10, so multicollinearity does not prevent multivariate estimation. It does, however, make individual coefficients less stable (Section 5.4).

5.2 Hypothesis tests (RQ1)

Table 4. Tests of H0: μ(distressed) = μ(healthy)

	Mean diff. t−1 [95% CI]	Paired t (wins.)	Paired p (raw)	Welch p	Wilcoxon p	Mann-Whitney p	Cohen's d	t−2 paired p	t−3 paired p
X1	−0.45 [−0.61, −0.30]	−5.54***	<0.001	<0.001	<0.001	<0.001	−2.24	0.009	0.190
X2	−0.42 [−0.54, −0.30]	−7.75***	<0.001	<0.001	<0.001	<0.001	−1.95	<0.001	<0.001
X3	−0.14 [−0.19, −0.10]	−6.31***	<0.001	<0.001	<0.001	<0.001	−2.62	<0.001	<0.001
X4	−11.63 [−30.23, −1.61]	−3.68***	0.219	0.219	<0.001	<0.001	−0.47	0.002	0.009
X5	−0.58 [−0.89, −0.28]	−3.87***	0.003	0.026	0.003	<0.001	−0.87	<0.001	<0.001
Z	−9.15 [−20.66, −2.90]	−6.16***	0.120	0.121	<0.001	<0.001	−0.60	<0.001	<0.001
Z"	−7.27 [−10.01, −4.82]	−6.81***	<0.001	<0.001	<0.001	<0.001	−2.08	<0.001	0.001

*Mean difference = distressed − healthy within each pair (bootstrap 95% CI, 5,000 resamples). Paired t (wins.) uses ratios winsorised at the pooled 5th/95th percentiles; *** p < 0.01, ** p < 0.05, * p < 0.10. The t−2 and t−3 columns report the winsorised paired-t p-value.*

H0 is rejected for every ratio at t−1. All within-pair differences are negative, as Altman's model predicts, and their bootstrap confidence intervals exclude zero. The effects are very large: Cohen's d is about −2 or larger for X1, X2, X3 and Z". The only disagreement between tests concerns the raw X4 and Z, whose paired t-tests are not significant (p = 0.219 and 0.120). The reason is not a weak effect. Two healthy peers have extreme market-to-liability ratios that inflate the variance of the differences. Once these are winsorised, or rank-based tests are used, the difference is highly significant (Wilcoxon p = <0.001). Shapiro-Wilk tests reject normality for

X4 and X5 in both groups, which is why the non-parametric tests are reported alongside the t-tests. The differences remain significant at t−2 for all ratios and at t−3 for every ratio except working capital ($p = 0.190$). Profitability and retained earnings separate the groups earliest, while liquidity deteriorates only in the last two years. The answer to RQ1 is therefore yes: all five Altman ratios differ significantly between future IBC companies and matched healthy firms, and four of the five already do so three years before the event.

5.3 Classification by the original Z and the Z''-score (RQ2)

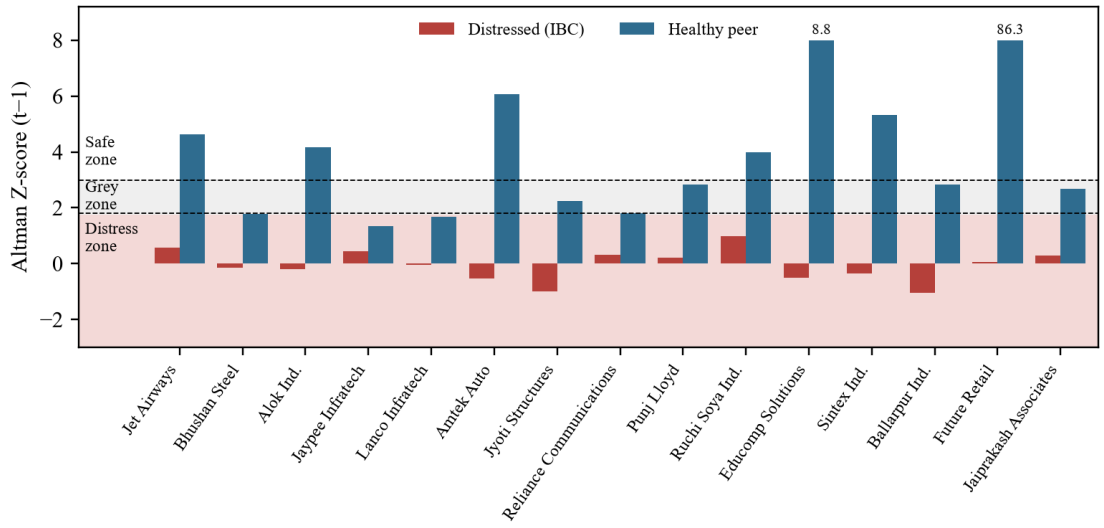


Figure 2. Z-score of each distressed firm and its matched peer at t−1 (values above 8 are truncated and labelled)

Table 5. Classification accuracy by horizon, compared with Altman (1968)

Rule	t−1 D hit	t−1 H hit	t−1 acc.	t−2 D hit	t−2 H hit	t−2 acc.	t−3 D hit	t−3 H hit	t−3 acc.
Z, cut-off 2.675	15/15	9/15	80%	15/15	7/14	76%	12/13	7/13	73%
Z, cut-off 1.81	15/15	11/15	87%	15/15	11/14	90%	11/13	12/13	88%
Z, cut-off 2.99	15/15	7/15	73%	15/15	6/14	72%	12/13	6/13	69%
Z'', cut-off 1.10	14/15	15/15	97%	11/15	13/14	83%	6/13	13/13	73%
Z'', cut-off 2.60	15/15	9/15	80%	14/15	12/14	90%	8/13	10/13	69%
Altman (1968), cut-off 2.675	**31/33**	**32/33**	**95%**	**72%**	–	–	**≈50%**	–	–

D hit = distressed firms correctly flagged (1 − Type I error); H hit = healthy firms correctly passed (1 − Type II error). 'Cut-off 1.81' treats the grey zone as healthy and 'cut-off 2.99' treats it as distressed. Altman's figures are for his original sample of US manufacturers; for t−2 and t−3 only the share of bankrupt firms correctly classified is shown.

Figure 2 and Table 5 give the main answer to RQ2. All 15 distressed firms were in the distress zone at t−1, so whichever cut-off is used, the Type I error, the costly error for an investor, is zero. Their Z-scores range from −1.05 (Ballarpur Industries) to 0.98 (Ruchi Soya), far below 1.81. The weakness lies on the other side. Of the healthy peers, only 7 scored above 2.99, 4 fell in the grey zone and 4 were in the distress zone: Sobha (1.35), Torrent Power (1.69), JSW Steel (1.77), Bharti Airtel (1.80). All four are capital-intensive, with low asset turnover and heavy fixed assets, a profile that Altman's 1968 weights, estimated on manufacturers of 1946–1965,

penalise. Grice and Ingram (2001) likewise found that the model is less accurate in recent samples and outside manufacturing. Bharti Airtel's FY2017 score also reflects the Jio price war, which pushed the telecom industry into losses that year. So with Altman's cut-offs, the overall accuracy of 87% at t−1 comes entirely from zero missed distressed firms combined with a Type II error of 27%.

The Z''-score, which drops the asset-turnover term and uses book equity, corrects much of this industry bias. No healthy peer falls in its distress zone (6 are grey), and with the 1.10 cut-off it classifies 97% of firms correctly at t−1, missing only Jaypee Infratech. At longer horizons, however, the Z''-score loses more power than the original Z (AUC 0.84 against 0.91 at t−3, Figure 3) because book equity responds more slowly than market value. The original Z with the 1.81 cut-off flagged 11 of 13 distressed firms three years ahead and misclassified only 1 healthy firm. That is far better than Altman's own long-range results (about half of bankrupt firms at t−3). The likely reason is specific to India: most of these companies had been non-performing assets for years before the IBC gave creditors an effective way to act. The 2017 RBI referral covered accounts that were 60% or more non-performing as of March 2016 (Reserve Bank of India, 2017). The admission date therefore comes late in a long period of distress, and the Z-score picks up that distress early.

5.4 Re-estimated discriminant function (RQ3)

Table 6. Multiple discriminant analysis re-estimated on the Indian t−1 sample (winsorised ratios)

Variable	Coefficient	Univariate F (1, 28)	Scaled coefficient	Rank by scaled coefficient
X1 – Working capital / Total assets	8.336	41.20***	1.51	3
X2 – Retained earnings / Total assets	15.800	39.78***	2.42	1
X3 – EBIT / Total assets	30.000	53.59***	1.58	2
X4 – Market value of equity / Total liabilities	−0.406	13.50***	−0.90	5
X5 – Sales / Total assets	2.187	6.56**	1.13	4

Higher scores indicate financial health; cut-off (midpoint of the group centroids) = 4.975. Wilks' $\lambda = 0.221$, Bartlett $\chi^2(5) = 38.53$, $p < 0.001$. Scaled coefficient = coefficient × pooled within-group standard deviation; ranks are by absolute scaled coefficient.

The re-estimated function is highly significant (Wilks' $\lambda = 0.221$, $\chi^2 = 38.5$, $p < 0.001$), and every ratio separates the groups on its own (Table 6). X3 has the largest univariate F-ratio, which is consistent with Altman's (1968) finding that earning power (X3) contributes most to the discriminant function. Asset turnover (X5) is the weakest, which mirrors Altman's 1968 sample, where X5 was not significant on its own. The multivariate weights differ from Altman's, though. In the Indian function, retained earnings (X2) carries the largest scaled weight, followed by profitability (X3) and working capital (X1), while X4 receives a small negative weight once the other ratios are controlled for. This sign reversal reflects its overlap with X2 and X3 (Section 5.1), not a genuine negative effect of market value. The ranking suggests that for Indian firms,

the depletion of accumulated profits, which captures years of losses and write-offs, is the most telling single symptom.

The function classifies 100% of the estimation sample correctly and 97% under leave-one-out cross-validation (one healthy firm misclassified). Applied without change to earlier years, it classifies 83% correctly at t−2 and 73% at t−3. Its Type I error rises to 27% and 54% because the function is tuned to the deep distress of the final year. On hold-out years, therefore, re-estimation does not outperform Altman's original weights with the 1.81 cut-off. This echoes Begley, Ming and Watts (1996), who found that these models lose accuracy outside the period in which they were estimated.

5.5 Logistic regression (RQ3)

Table 7. Logit models of the probability of distress at t−1 (Firth penalised likelihood)

	Variable	Coefficient	Std. error	p-value	Odds ratio
Model 1	Constant	4.264	2.016	0.034	–
	Z	−3.722	1.635	0.023	0.024
	McFadden R^2 = 0.928; LR $\chi^2(1)$ = 38.58, $p < 0.001$; AIC = 7.00				
Model 2	Constant	3.158	1.917	0.099	–
	X1	−0.848	4.102	0.836	0.428
	X2	−10.056	6.131	0.101	0.000
	X3	−25.280	19.738	0.200	0.000
	X4	0.520	0.437	0.234	1.682
	X5	−1.380	1.558	0.376	0.252
	McFadden R^2 = 0.861; LR $\chi^2(5)$ = 35.80, $p < 0.001$; AIC = 17.79				

Dependent variable = 1 if the firm was later admitted to CIRP. Maximum likelihood fails because of complete separation, so Firth (1993) bias-reduced estimates are reported. Model 2 uses winsorised ratios.

Maximum-likelihood logit estimation breaks down because the Z-score separates the two groups perfectly at t−1, so the Firth penalised estimator is used. In Model 1, the Z-score is a significant predictor of distress (coefficient −3.72, p = 0.023). Each one-point increase in Z multiplies the odds of later insolvency by 0.024, and the model's McFadden R^2 is 0.93. Model 2 is jointly highly significant (LR χ^2 = 35.8, $p < 0.001$), but no single ratio is individually significant, and its AIC (17.8) is worse than Model 1's (7.0). With 30 observations and five correlated ratios, the data cannot separate the individual effects, whereas the composite Z summarises them efficiently. Model 2's leave-one-out accuracy is 97%, the same as the re-estimated MDA. In a small-sample screening setting, Altman's composite score therefore does as well as a freshly estimated set of weights, and Model 1 gives it the probability interpretation that Ohlson (1980) advocated.

5.6 Model comparison, optimal cut-off and robustness

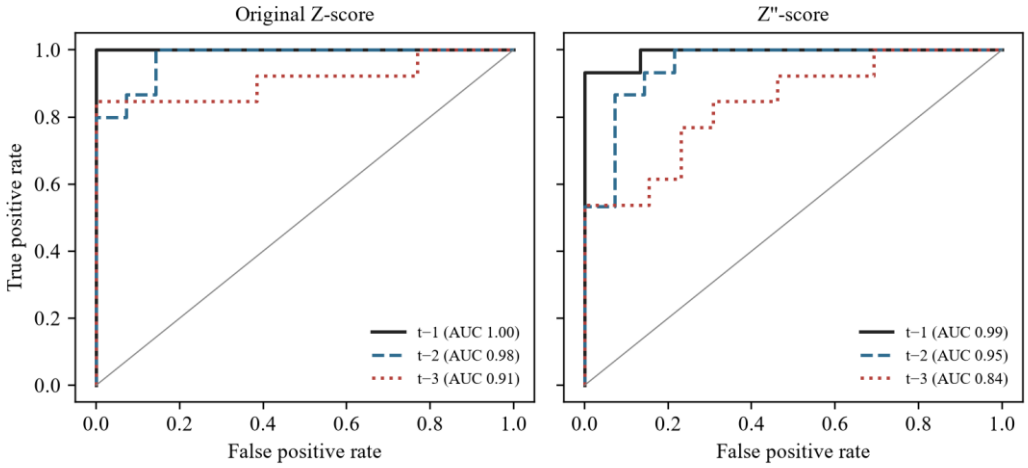


Figure 3. ROC curves of the original Z and the Z"-score at t−1, t−2 and t−3

Table 8. Area under the ROC curve at t−1 (re-estimated models scored out of sample)

Model	AUC	Bootstrap 95% CI
Altman Z (1968)	1.000	[1.000, 1.000]
Altman Z" (emerging market)	0.991	[0.960, 1.000]
Re-estimated MDA (LOO)	0.996	[0.973, 1.000]
Logit X1-X5 (LOO)	0.991	[0.956, 1.000]

At t−1 all four models discriminate almost perfectly (AUC ≥ 0.99, Table 8), and DeLong tests find no significant difference between any pair (smallest p = 0.409). The re-estimated MDA and logit therefore add nothing to the original Z-score in ranking firms by risk. What re-estimation does show is that Altman's cut-offs are set too high for Indian firms. The Youden-optimal cut-off for the original Z lies anywhere between 0.98 and 1.35, so we take its midpoint, 1.17. In sample it classifies every firm correctly. More importantly, when it is applied out of time, it classifies 90% correctly at t−2 and 85% at t−3, without flagging a single healthy firm at t−3. This is in line with the evidence of Bandyopadhyay (2006) and Singh and Mishra (2016) that recalibrating the model to local data improves accuracy for Indian firms. It is a cut-off for this sample, though, and needs validation on a larger one before being used as a rule.

Table 9. Robustness of the t−1 results

Specification	n	Accuracy	Type I	Type II	AUC
Baseline: original Z, market equity (cut-off 2.675)	30	80%	0%	40%	1.000
Book equity in X4 (cut-off 2.675)	30	73%	0%	53%	1.000
Excluding airline and telecom pairs (cut-off 2.675)	26	81%	0%	38%	1.000
Pairs with LSEG-sourced working capital only (cut-off 2.675)	16	75%	0%	50%	1.000
Original Z with the single cut-off 1.81	30	87%	0%	27%	1.000
Original Z with the single cut-off 2.99	30	73%	0%	53%	1.000
Z" (book equity), cut-off 1.10	30	97%	7%	0%	0.991

Table 9 shows that the conclusions do not depend on how the variables are measured or which industries are included. Using book instead of market equity in X4 leaves the AUC unchanged but raises the Type II error, because the market had already marked distressed equity down.

Excluding the airline and telecom pairs, whose 2017–2019 industry shocks could drive the result, leaves accuracy essentially unchanged, and restricting the sample to pairs whose working capital comes entirely from the LSEG data gives the same picture. Across every specification of the original Z, the Type I error at t−1 is zero, and the remaining errors are healthy firms flagged as distressed.

5.7 Investment implications (RQ4)

Table 10. Z-score path of the distressed firms and share-price change to the admission date

Distressed company	Z t−3	Z t−2	Z t−1	CIRP admission	Share return	Peer return
Jet Airways (India)	0.64	1.16	0.58	20 Jun 2019	-87%	28%
Bhushan Steel	0.21	−0.06	−0.15	26 Jul 2017	13%	14%
Alok Industries	–	0.19	−0.19	18 Jul 2017	-5%	-12%
Jaypee Infratech	1.00	0.91	0.45	09 Aug 2017	98%	12%
Lanco Infratech	0.16	0.29	−0.05	07 Aug 2017	-64%	-19%
Amtek Auto	–	0.26	−0.54	24 Jul 2017	-11%	9%
Jyoti Structures	1.18	0.54	−0.99	04 Jul 2017	25%	24%
Reliance Communications	1.20	0.75	0.33	15 May 2018	-64%	10%
Punj Lloyd	0.42	0.31	0.21	08 Mar 2019	-87%	-17%
Ruchi Soya Industries	2.39	1.75	0.98	15 Dec 2017	-29%	-2%
Educomp Solutions	1.03	−0.30	−0.52	30 May 2017	-25%	227%
Sintex Industries	0.99	0.82	−0.35	06 Apr 2021	438%	217%
Ballarpur Industries	−0.49	−0.25	−1.05	17 Jan 2020	-85%	-7%
Future Retail	4.99	1.58	0.05	20 Jul 2022	-85%	38%
Jaiprakash Associates	0.47	0.31	0.29	03 Jun 2024	122%	-2%

Share return = change in the NSE closing price (adjusted for corporate actions) from the last trading day of fiscal year t−1 to the last trading day before the admission order; the peer is measured over the same window.

For an investor, the key question is how early the warning comes and what ignoring it would cost. Table 10 shows that the warning came early. Of the 13 distressed firms with data for t−3, 11 were already in the distress zone three years before admission, and all 15 were there by t−2. The exceptions, Ruchi Soya and Future Retail, are companies whose distress set in quickly: commodity-margin losses at Ruchi, and Future Retail's collapse during the COVID-19 lockdowns. Between the end of year t−1 and the admission order, the median distressed share fell 25% while the median peer gained 10%. Jet Airways, Punj Lloyd, Ballarpur Industries and Future Retail lost more than 80%. A few low-priced shares (Jaypee Infratech, Sintex Industries, Jaiprakash Associates) rose sharply on speculation about resolution bids. This is a reminder that a distress signal measures the risk of default, not short-run price movements, which agrees with Campbell, Hilscher and Szilagyi's (2008) finding that distressed stocks deliver poor and highly volatile returns.

For RQ4, the practical rule is simple: a Z below about 1.2, or a Z" below 1.10, should exclude an Indian company from a long-only equity or unsecured-credit portfolio unless deeper analysis shows otherwise. A score between that level and 3.0 should prompt an industry-adjusted review

rather than an automatic sale, especially in steel, power, real estate and telecom, where sound firms naturally score low.

6. Conclusion

This project tested Altman's (1968) Z-score on fifteen Indian listed companies admitted to insolvency under the IBC between 2017 and 2024, each matched with a healthy listed peer, using standalone statements cross-checked across two independent data compilations and against annual reports. The main findings are:

(1) All five Altman ratios are significantly lower for distressed firms one and two years before admission, and all except working capital already differ three years before (RQ1). Retained earnings and operating profitability separate the groups earliest; liquidity collapses last.

(2) Every distressed company was in Altman's distress zone a year before admission (Type I error 0%), and 11 of 13 were already there three years before. The original model's weakness for Indian firms is false alarms: 4 of 15 healthy, capital-intensive peers scored below 1.81, giving overall accuracy of 87% at t−1. The emerging-market Z"-score reduces this industry bias and reaches 97% accuracy at t−1, but it loses power faster at longer horizons (RQ2).

(3) Re-estimating the model on Indian data does not improve the ranking of firms by risk (all AUCs $\geq$ 0.99; DeLong tests not significant). It does show that retained earnings carry the most weight for Indian firms and that a lower cut-off of about 1.2 removes most false alarms while keeping its early-warning power out of time (RQ3).

(4) For investors, the Z-score gives a cheap and early warning. Distressed shares lost a median 25% between the end of year t−1 and admission, while their peers gained 10% (RQ4).

Limitations. The matched-pair sample is small (30 firms) and selected on the outcome, which Zmijewski (1984) shows overstates accuracy; the in-sample cut-off in particular needs validation on a larger, unmatched population before it is used as a rule. The distressed firms were also deeply distressed by the time of admission, so the near-perfect t−1 separation partly reflects how late the IBC event occurs rather than how sensitive the model is. Retained earnings are measured as reserves and surplus, and share prices for some distressed firms were thinly traded. Future work could use the full population of listed non-financial firms with a hazard model (Shumway, 2001), add market-based measures such as distance to default (Hillegeist et al., 2004), and estimate industry-specific cut-offs for India.

Overall, more than fifty years after it was proposed, Altman's Z-score remains a dependable first screen of financial health for Indian companies. It rarely misses a company heading for insolvency, but in capital-intensive industries its cut-offs should be lowered or the Z''-score used instead, so that sound firms are not flagged as distressed.

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Appendix: Sample Firms and Event Dates

Table A1. Matched sample, event dates and sources of the admission date

#	Distressed company	Healthy peer	Industry	CIRP admitted	Years (t−3 to t−1)	Source of admission date
1	Jet Airways (India)	InterGlobe Aviation	Airlines	20 Jun 2019	FY2016–FY2018	NCLT Mumbai order on SBI petition (Business Today, 20 Jun 2019)
2	Bhushan Steel	JSW Steel	Integrated steel	26 Jul 2017	FY2015–FY2017	NCLT Principal Bench order on SBI petition (Business Today, 26

						Jul 2017)
3	Alok Industries	Vardhman Textiles	Textiles	18 Jul 2017	FY2015–FY2017	NCLT Ahmedabad order on SBI petition (Business Standard, 19 Jul 2017)
4	Jaypee Infratech	Sobha	Real estate & infrastructure	09 Aug 2017	FY2015–FY2017	NCLT Allahabad order on IDBI Bank petition (Business Standard, 10 Aug 2017)
5	Lanco Infratech	Torrent Power	Power generation & EPC	07 Aug 2017	FY2015–FY2017	NCLT Hyderabad order on IDBI Bank-led petition (BloombergQuint, 9 Aug 2017)
6	Amtek Auto	Bharat Forge	Auto components (forgings)	24 Jul 2017	FY2015–FY2017	NCLT Chandigarh order on Corporation Bank petition (BSE filing, 24 Jul 2017)
7	Jyoti Structures	KEC International	Power transmission EPC	04 Jul 2017	FY2015–FY2017	NCLT Mumbai order on SBI petition (Hindustan Times, 5 Jul 2017)
8	Reliance Communications	Bharti Airtel	Telecommunications	15 May 2018	FY2015–FY2017	NCLT Mumbai order on Ericsson India petition (Business Standard, 16 May 2018)
9	Punj Lloyd	Kalpataru Power Transmission	Engineering, procurement & construction	08 Mar 2019	FY2016–FY2018	NCLT Principal Bench order on ICICI Bank petition (company filing; IBBI)
10	Ruchi Soya Industries	Gokul Agro Resources	Edible oils	15 Dec 2017	FY2015–FY2017	NCLT Mumbai order on Standard Chartered/DBS petitions (Business Standard; IBBI)
11	Educomp Solutions	Aptech	Education services	30 May 2017	FY2014–FY2016	NCLT Principal Bench order on Section 10 application (company filing; IBBI)
12	Sintex Industries	K.P.R. Mill	Textiles	06 Apr 2021	FY2018–FY2020	NCLT Ahmedabad order on Invesco AMC petition, CP(IB) 848/2019 (IBBI)
13	Ballarpur Industries	JK Paper	Paper	17 Jan 2020	FY2017–FY2019	NCLT Mumbai order on Finquest Financial Solutions petition, CP 2915/2019 (company annual report 2019-20; IBBI)
14	Future Retail	Avenue Supermarts	Organised retail	20 Jul 2022	FY2019–FY2021	NCLT Mumbai order on Bank of India petition (Business Standard, 20 Jul 2022)
15	Jaiprakash Associates	The Ramco Cements	Cement & construction	03 Jun 2024	FY2021–FY2023	NCLT Allahabad order on ICICI Bank and SBI petitions (LiveLaw, Jun 2024)

FY = fiscal year ending 31 March. Future Retail's t−1 is FY2021 because its FY2022 results were delayed (company filing of 24 May 2022) and no FY2022 statements are available in the data sources. Ratios, raw figures and the source of every working-capital figure are listed in Z_Score_Dataset.xlsx.